

Department of Information and Communication Technology
Faculty of Technology
University of Ruhuna

Assignment – Practical

Duration: 60 min

ICT1253 – Computer Networks

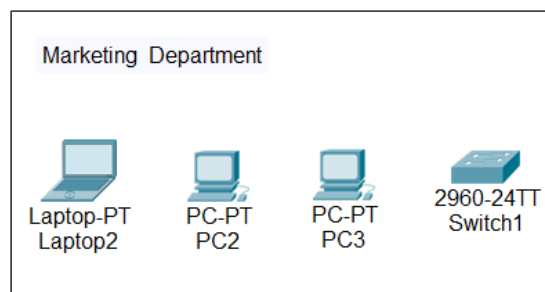
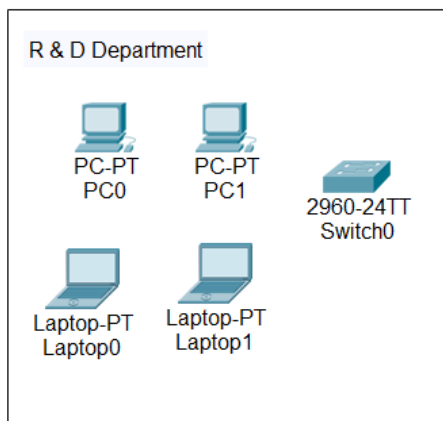
Instructions

- You have to create the configured network for the question and save your .pkt file as **Index_Number.pkt**

NOTE: After finishing the question, you have to save your configurations to **startup configuration** before saving your document.

- You have to answer the questions on Microsoft Word Document and save it as **Index_Number.doc or Index_Number.docx**. It must include configuration details in plain text. Photos, Images or screenshots of configuration details are not accepted.
- Submit both .pkt file and word document as zip file named with your Index Number.

Question: Look at the following image and given description. Answer questions below.



There is a small organization named “ABC Ptv Ltd” that has two departments called Research and Development (R&D) and Marketing. Two departments are located in two floor of the same building and each department has their own network. Communication between two departments is done with the aid of a **2911 Router**.

1. What is the functionality of a router and a switch?
2. Open Cisco Packet Tracer and add above equipment. What type of cable(s) that we have to use in this network?
3. Arrange devices and connect them with proper cables.
4. Assign these relevant IP addresses to the end devices

Device Name	IP Address
Laptop0	200.165.10.1
Laptop1	200.165.10.2
PC0	200.165.10.3
PC1	200.165.10.4
Laptop2	200.165.20.1
PC2	200.165.20.2
PC3	200.165.20.3

5. The subnet mask of all seven (07) devices are 255.255.255.0.

Switch0 is connected to Router's GigabitEthernet0/0 interface while the Switch1 is connected to Router's GigabitEthernet0/1 interface.

6. Change the router's IP addresses as follows

Interface	IP Address
GigabitEthernet0/0	200.165.10.10
GigabitEthernet0/1	200.165.20.10
GigabitEthernet0/2	200.165.30.10

7. Activate or up the connection of both interfaces on the router.
8. Assign the appropriate IP address of the Router's interfaces as the default gateway of each end device connected to the switches.
(Hint: Laptop0 is connected to Switch0 and Switch0 is accessing router through GigabitEthernet0/0 interface)

In the near future apart from the existing 2 departments, 3 more departments are to be created in the "ABC Pvt Ltd" organization. Departments to be newly added in the future will also maintain their own network. By considering all 5 departments optimal subnets are required to be generated.

9. Change the subnet mask of all end devices and the router considering the above requirement.

10. Ping in between different networks.

- i. Ping from PC0 to Laptop2
- ii. Ping from PC3 to PC1
- iii. Ping from Laptop0 to Laptop1

Is all successful or not? (Add screenshots of the command prompt where pinging is done)

NOTE: Screenshots are accepted for this part (Part 10) only.

11. Configure Switch0 as follow and copy all the configuration done in the CLI to your word document answer sheet as plain text.

- a. Assign name "S0" to Switch0
- b. Examine the current content running configuration file
- c. Configure the vty lines 0 15 with the password "cisco" and enable login
- d. Configure the con 0 with the password "cisco" and enable login
- e. Examine the content save in NVRAM
- f. Save all the running configurations to NVRAM
- g. Show the content in flash storage

12. Configure the Router as follow and copy all the configuration done in the CLI to your word document answer sheet as plain text.

- a. Assign name "R1" to the Router
- b. Disable DNS lookup
- c. Set the minimum password length to 8 characters
- d. Assign "ciscopwdnew" as the privilege execution mode password
- e. Assign "ciscovtypwd" as the vty 0 4 password with a timeout of 6 minutes 20 seconds and enable login and enable logging synchronous
- f. Encrypt all clear text passwords
- g. Create a banner, "Unauthorized Access Prohibited" that warns anyone accessing the device
- h. Save all the running configurations to NVRAM

13. What is the purpose of "logging synchronous" command?

14. Add a new PC and provide it with IP address 200.165.30.1 with the optimal subnet mask found in part 09.

15. Connect the newly added PC directly to the Router's GigabitEthernet0/2 interface and set the default gateway of the new PC appropriately.

16. Remotely access the Router from **newly added PC** using **telnet** protocol and copy the below given commands to be run in the console of the router remotely from PC2 to your work document answer sheet as a plain text.
- a. Display the version of the router
 - b. Filter the content (size) regarding the non-volatile random-access memory the router have.